

# Dr. Romain Mottier

Computational Scientist — PhD in Applied Mathematics

Basel region | romain.mottier@outlook.com | +33 6 23 23 24 29 | finiteclarity.com
romainmottier.github.io | github.com/romainmottier | linkedin.com/in/romain-mottier

French (native) • English (C2) • French citizen – Swiss work permit (G)

## PROFILE

**Computational Scientist in Applied Mathematics** & **Physics.** Five years developing and shipping simulation code for engineering and physics problems: wave propagation, solid–fluid interaction, geophysics and electromagnetics. Production-level **C++ and Fortran**, Python for scripting and post-processing, from **PDE solver design** and mathematical validation to large-scale parallel runs (**MPI/OpenMP**) on HPC clusters. Contributor to **SEM3D**, a production seismic simulation code used for nuclear-site studies.

**Position sought.** R&D and scientific engineering roles where rigour translates into industrial applications. Comfortable entering a new physical or mathematical domain from scratch: four distinct fields in five years, each carried through to working, validated code.

## PROFESSIONAL EXPERIENCE

**Freelance Consultant — Scientific Computing** 10/2026 – present
Finite Clarity (finiteclarity.com) Basel region

- Consulting and development in scientific computing for R&D teams in industry and research.
- Code diagnostics, **verification and validation**, modernisation and CPU/GPU parallelisation, custom solvers and modules for in-house codes or commercial software.
- On-demand simulation studies with proof of convergence (flows, structures, heat transfer, waves, coupled physics), part-time expert support, training in numerical analysis and HPC.

**Postdoctoral Researcher — Computational Mathematics** 10/2025 – present
University of Basel, Dept. of Mathematics and Computer Science Basel, Switzerland

- Extending **DiSk++** (C++17 finite element library) with multilevel explicit local time-stepping, so a locally refined mesh no longer forces the whole domain down to the smallest time step.
- Developing high-order solvers for strongly nonlinear PDEs whose nonlinearity varies in space.
- Presented at **five conferences and workshops** in 2025–2026 (Delft, Vienna, Montreal, Zurich, Turin); two preprints on arXiv, one article in preparation.
- Sole lecturer for a final-year BSc course in numerical analysis of ODEs (Fall 2026)
- Supervised **about ten Master's students** on semester projects in numerical simulation (finite elements for linear elasticity; spectral-element solver for acoustic waves), from project definition to final report.

**PhD Researcher — Applied Mathematics & Scientific Computing** 10/2021 – 07/2025
French Atomic Energy Commission (CEA) & École nationale des Ponts et Chaussées (ENPC) Paris, France
& Polytechnic Institute of Paris (IP Paris)

- Developed and analysed high-order finite element methods for coupled **elasto-acoustic (solid–fluid) wave propagation**, carried from mathematical analysis to validated implementations in C++ and Fortran.
- Ran 3D seismic wave simulations on the **CEA EXA1 supercomputer**: tens of millions of unknowns on 1,000+ cores (MPI/OpenMP).
- Worked across three institutions, reporting to both academic supervisors and CEA industrial researchers.
- Authored **3 research articles** published in leading international journals and presented at **5 international and national conferences**.
- **Python** lab sessions in numerical methods at Paris Dauphine and Sorbonne University (BSc and MSc).

**Research Intern — Computational Electromagnetics** 03/2021 – 08/2021
ONERA, The French Aerospace Lab Toulouse, France

- Implemented Compatible Discrete Operator (CDO) schemes for the time-domain Maxwell equations, benchmarked against reference solutions (Fortran 90).

**Research Intern — Planetary Thermal Modelling** 05/2020 – 08/2020
European Space Agency (ESA/ESTEC) Noordwijk, Netherlands

- Implemented a heat-transfer model of the surface and sub-surface temperature of Mercury (Fortran 90).

## SOFTWARE PROJECTS

**DiSk++** github.com/wareHHOuse/diskpp

C++17 template library for discontinuous skeletal methods (HHO/HDG)

- Contributed the **elasto-acoustic wave propagation module** to an open-source scientific library. Implemented templated C++ covering assembly, static condensation and explicit/implicit time integration, dimension- and mesh-agnostic by design — supporting the results published in **M2AN** (2025) and **CMAME** (2026).
- Built on Eigen, MUMPS, SILO/HDF5, Gmsh and CMake; validated against analytical solutions on 2D/3D benchmarks up to order 7, with problems of several million degrees of freedom.

**SEM3D** github.com/sem3d/SEM

Fortran 90 spectral-element solver for 3D seismic wave propagation

- Implemented the HDG method for the elastodynamic wave equation, with explicit time integration and MPI/OpenMP parallelisation, in a **production geophysics code** developed by CEA and academic partners for broadband (0–10 Hz) regional-scale seismic simulation at nuclear sites.

**ProtoN** github.com/wareHHOuse/ProtoN

C++ library for prototyping cut/unfitted finite element methods

- Implemented discontinuous cut-FEM (unfitted HHO) schemes with polynomial-extension stabilisation for elliptic interface problems, supporting the results published in **SINUM** (2026).

## TECHNICAL SKILLS

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| **Programming** | **C++** • **Python** • **Fortran** • MATLAB • Bash |
| **Applied mathematics** | Partial Differential Equations • Numerical Analysis • Linear Algebra • Optimisation |
| **Scientific computing** | High-Performance Computing (MPI, OpenMP) • Eigen • Intel MKL • MUMPS • Gmsh |
| **Numerical methods** | Finite Element & Finite Volume Methods • Time Integration • Sparse Linear Solvers |
| **Data & analysis** | NumPy • Pandas • Matplotlib • ParaView |
| **Software engineering** | Git • CMake • Unit testing • Code review • Linux • LLM coding assistants (Claude Code, Copilot) |
| **Application domains** | Wave Propagation & Solid–Fluid Interaction • Geophysics • Electromagnetics |

## EDUCATION

**PhD in Applied Mathematics & Scientific Computing** 10/2021 – 07/2025
French Atomic Energy Commission (CEA) & École nationale des Ponts et Chaussées (ENPC) Paris, France
& Polytechnic Institute of Paris (IP Paris)

- Thesis: *Hybrid high-order methods for the numerical simulation of elasto-acoustic wave propagation*
  Defended July, 2025, supervised by A. Ern and L. Guillot.

**MSc in Engineering: Modelling and Fluid–Structure Computation** 09/2018 – 09/2021
Université de Toulon, SeaTech Engineering School Toulon, France

- Exchange semester — *MSc Numerical Methods in Engineering*,
  Universitat Politècnica de Catalunya (UPC), Barcelona, Spain (09/2020 – 02/2021).

## PUBLICATIONS & CONFERENCES

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| **In prep.** | Multilevel explicit and locally implicit Runge–Kutta time stepping for hybrid high-order methods for wave equations |
| **arXiv 2026** | A Hybrid High-Order method for variable-exponent Leray–Lions problems with convergence under minimal regularity |
| **arXiv 2026** | A priori error estimates for a Hybrid High-Order method on variable-exponent Leray–Lions problems |
| **SINUM 2026** | Unfitted hybrid high-order methods stabilized by polynomial extension for elliptic interface problems |
| **CMAME 2026** | Elasto-acoustic wave propagation in geophysical media using hybrid high-order methods on general meshes |
| **M2AN 2025** | Hybrid high-order methods for elasto-acoustic wave propagation in the time domain |

Co-authors: M. J. Grote (in preparation) • A. Harnist (preprints) • E. Burman, A. Ern, L. Guillot, R. Khot (journal articles).

**Talks** — Wave-NED (Delft, 08/2026) • CMAM (Vienna, 07/2026) • WAVES (Montreal, 06/2026) • Swiss Numerics Day (Zurich, 04/2026) • PoliTopes workshop (Turin, 12/2025) • WCCM (Vancouver, 07/2024) • ECCOMAS (Lisbon, 06/2024) • CANUM (Île de Ré, 05/2024).

**Posters** — AGU Fall Meeting (San Francisco, 12/2023) • CJCMA (Paris, 09/2023).

## REFERENCES

**Contact details available on request**

**Prof. Marcus J. Grote** — Postdoctoral advisor
Professor, University of Basel

**Prof. Alexandre Ern** — PhD advisor
Senior Researcher, INRIA (SERENA)

**Dr. Laurent Guillot** — Industrial PhD advisor
Researcher, CEA

**Dr. Guillaume Delay**
Researcher, Sorbonne University (LJLL)